



# ITESO

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Proyecto FINAL - MPLS

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# MPLS VPN Final Practice Report

## 1. Introduction

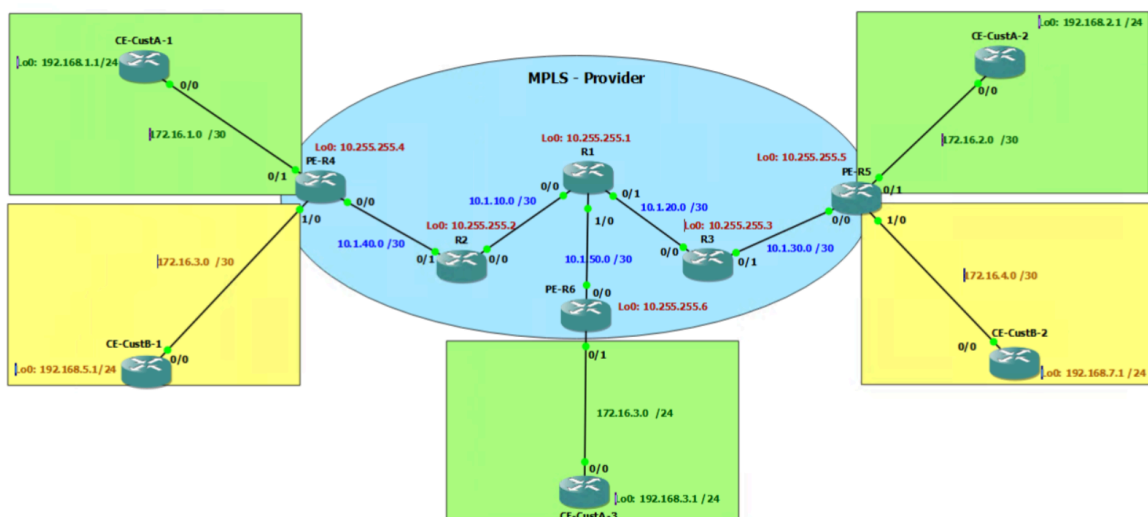
The purpose of this project was to design and implement a fully functional MPLS VPN network topology using Cisco devices in a simulated lab environment. The primary objective was to establish a provider infrastructure capable of offering private, isolated communication for multiple customers—specifically, CustomerA and CustomerB—by using MPLS, BGP, and VRFs (Virtual Routing and Forwarding). Each customer was to be strictly confined to their own routing table, with no interaction between them.

The backbone network had to demonstrate full route convergence using an interior gateway protocol (EIGRP), and data forwarding had to be managed through MPLS labels. We also had to ensure successful BGP peering between the provider edge (PE) and customer edge (CE) routers, validate data flow using ping and traceroute commands (ensuring label usage and hop concealment), and enforce device-level security using encrypted login credentials. This was aimed at reinforcing advanced concepts in service provider networking, secure access, and traffic isolation.

## 2. Theoretical Framework

MPLS (Multiprotocol Label Switching) is a high-performance technique that directs data based on short path labels rather than long IP addresses, which accelerates packet forwarding. BGP (Border Gateway Protocol) is the de facto standard for routing between autonomous systems and is crucial for distributing VPNv4 routes in an MPLS VPN architecture. VRFs are logical instances that maintain separate routing tables on the same router, enabling isolation between customers. EIGRP (Enhanced Interior Gateway Routing Protocol) serves as the provider's IGP, used for fast convergence and route distribution between loopback and transit interfaces. Together, these technologies allow scalable, secure, and isolated communication in a shared infrastructure.

### Topology diagram:



## 3. Configuration Steps

### 3.1 Core Routers Configuration (R1, R2, R3)

These routers operate solely in the provider's core. They are configured with MPLS for label distribution and EIGRP for internal convergence. Each router uses its loopback interface as a unique identifier.

*Enabled MPLS globally and set label protocol:*

```
mpls ip
mpls label protocol ldp
```

*Loopback interface as the LDP router ID:*

```
interface Loopback0
ip address 10.255.255.X 255.255.255.255
no shutdown
mpls ldp router-id Loopback0
```

*Inter-router interfaces with IP addresses and MPLS:*

```
interface f0/0
ip address 10.1.10.X 255.255.255.252
no shutdown
mpls ip
```

```
interface f0/1
ip address 10.1.20.X 255.255.255.252
no shutdown
mpls ip
```

*EIGRP configuration to advertise the loopback and core links:*

```
router eigrp 100
network 10.255.255.X 0.0.0.0
network 10.1.10.0 0.0.0.3
network 10.1.20.0 0.0.0.3
no auto-summary
```

### **3.2 PE Routers Configuration (PE-R4, PE-R5, PE-R6)**

PE routers interact with both core and customer routers. Each is configured with MPLS, VRFs for CustomerA and/or CustomerB, EIGRP in the core, and BGP sessions for both internal (VPNv4) and external (CE-facing) communication.

*Enabled MPLS globally and defined LDP behavior:*

```
mpls ip
mpls label protocol ldp
```

*Loopback as router ID and enable MPLS on core-facing interfaces:*

```
interface Loopback0
```

```
ip address 10.255.255.X 255.255.255.255
no shutdown
mpls ldp router-id Loopback0
```

```
interface f0/0
ip address 10.1.XX.X 255.255.255.252
no shutdown
mpls ip
```

#### *Customer-facing interfaces to VRFs and IPs:*

```
interface f0/1
ip vrf forwarding customerA
ip address 172.16.1.1 255.255.255.252
no shutdown
```

```
interface f1/0
ip vrf forwarding customerB
ip address 172.16.2.1 255.255.255.252
no shutdown
```

#### *VRFs and route-targets:*

```
ip vrf customerA
rd 100:1
route-target export 100:1
route-target import 100:1
```

```
ip vrf customerB
rd 200:1
route-target export 200:1
route-target import 200:1
```

#### *Configure EIGRP for core interfaces:*

```
router eigrp 100
network 10.255.255.X 0.0.0.0
network 10.1.XX.0 0.0.0.3
no auto-summary
```

#### *BGP configuration for VPNv4 exchange and CE connectivity:*

```
router bgp 400
neighbor 10.255.255.Y remote-as 400
neighbor 10.255.255.Y update-source Loopback0
```

```
address-family vpnv4
neighbor 10.255.255.Y activate
neighbor 10.255.255.Y send-community both
neighbor 10.255.255.Y next-hop-self
```

```
address-family ipv4 vrf customerA
redistribute connected
neighbor 172.16.1.2 remote-as 65100
neighbor 172.16.1.2 activate
```

```
address-family ipv4 vrf customerB
redistribute connected
neighbor 172.16.2.2 remote-as 65100
neighbor 172.16.2.2 activate
```

### **3.3 CE Routers Configuration (CustomerA/B)**

Each customer router connects to a PE router via a single physical interface. These routers are not running MPLS or EIGRP but rely solely on BGP for advertisement.

*Configure loopback and WAN interface:*

```
interface Loopback0
ip address 192.168.X.1 255.255.255.0

interface f0/0
ip address 172.16.X.2 255.255.255.252
no shutdown
```

*Static default route to the PE router:*

```
ip route 0.0.0.0 0.0.0.0 172.16.X.1
```

*Configuration of BGP to advertise local networks and peer with the PE:*

```
router bgp 65100
network 192.168.X.0 mask 255.255.255.0
network 172.16.X.0 mask 255.255.255.252
neighbor 172.16.X.1 remote-as 400
no auto-summary
```

#### 4. Verification and Testing

- **EIGRP Convergence:** Verified with show ip eigrp neighbors, showing full adjacency between core and PE routers.
- **Label Assignment:** show mpls forwarding-table showed label bindings to destination prefixes.
- **VPN Isolation:** Ping tests confirmed that CustomerA could not reach CustomerB networks and vice versa.
- **MPLS Traceroute:** Using traceroute vrf customerA <destination IP> revealed hops with the respective label information; intermediate PE/P routers remained hidden.
- **Security Enforcement:** Encrypted login and password access were configured to secure administrative access.

```
service password-encryption
enable secret cisco123
username admin privilege 15 secret admin
line vty 0 4
login local
```

This ensures all access credentials are encrypted and authentication is required for CLI sessions.

#### 5. Evidence

```
R1(config)#do show ip eigrp ne
IP-EIGRP neighbors for process 100
H   Address                Interface      Hold Uptime    SRTT   RTO  Q  Seq
                               (sec)          (ms)          Cnt  Num
1   10.1.10.2               Fa0/0         14 01:12:29  1037  5000  0  29
2   10.1.50.2               Fa1/0         12 01:17:10   62    372  0  21
0   10.1.20.2               Fa0/1         14 01:17:11   75    450  0  35
```

R1 eigrp neighbors

```

R2(config)#do show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

    10.0.0.0/8 is variably subnetted, 11 subnets, 2 masks
C       10.1.10.0/30 is directly connected, FastEthernet0/0
C       10.255.255.2/32 is directly connected, Loopback0
D       10.255.255.3/32 [90/435200] via 10.1.10.1, 01:15:09, FastEthernet0/0
D       10.255.255.1/32 [90/409600] via 10.1.10.1, 01:15:09, FastEthernet0/0
D       10.255.255.6/32 [90/412160] via 10.1.10.1, 01:15:09, FastEthernet0/0
D       10.255.255.4/32 [90/409600] via 10.1.40.1, 01:19:46, FastEthernet0/1
D       10.255.255.5/32 [90/460800] via 10.1.10.1, 01:15:12, FastEthernet0/0
D       10.1.30.0/30 [90/332800] via 10.1.10.1, 01:15:12, FastEthernet0/0
D       10.1.20.0/30 [90/307200] via 10.1.10.1, 01:15:12, FastEthernet0/0
C       10.1.40.0/30 is directly connected, FastEthernet0/1
D       10.1.50.0/30 [90/284160] via 10.1.10.1, 01:15:12, FastEthernet0/0

```

## R2 ip route

```

PE-R4(config)#do show mpls forwarding-table
Local   Outgoing   Prefix          Bytes tag  Outgoing     Next Hop
tag      tag or VC   or Tunnel Id    switched  interface
16       Pop tag     10.255.255.2/32  0          Fa0/0        10.1.40.2
17       17          10.1.50.0/30     0          Fa0/0        10.1.40.2
18       18          10.255.255.1/32  0          Fa0/0        10.1.40.2
19       21          10.255.255.6/32  0          Fa0/0        10.1.40.2
20       16          10.1.20.0/30     0          Fa0/0        10.1.40.2
21       19          10.1.30.0/30     0          Fa0/0        10.1.40.2
22       Pop tag     10.1.10.0/30     0          Fa0/0        10.1.40.2
23       20          10.255.255.3/32  0          Fa0/0        10.1.40.2
24       22          10.255.255.5/32  0          Fa0/0        10.1.40.2
25       Aggregate  172.16.1.0/30[V] 4738
26       Untagged   192.168.1.0/24[V] 0          Fa0/1        172.16.1.2
27       25          172.16.2.0/30[V] 0          Fa0/0        10.1.40.2
28       25          172.16.3.0/30[V] 0          Fa0/0        10.1.40.2
29       26          172.16.3.0/24[V] 0          Fa0/0        10.1.40.2
30       26          192.168.2.0/24[V] 0          Fa0/0        10.1.40.2
31       27          192.168.3.0/24[V] 0          Fa0/0        10.1.40.2
32       27          172.16.4.0/30[V] 0          Fa0/0        10.1.40.2
33       28          192.168.7.0/24[V] 0          Fa0/0        10.1.40.2

```

## PE-R4 mpls forwarding-table



```

PE-R4(config)#do show ip bgp vpnv4 all
BGP table version is 32, local router ID is 10.255.255.4
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
               r RIB-failure, S Stale
Origin codes: i - IGP, e - EGP, ? - incomplete

   Network                Next Hop           Metric LocPrf Weight Path
Route Distinguisher: 100:1 (default for vrf customerA)
* 172.16.1.0/30           172.16.1.2             0             0 65100 i
*> 0.0.0.0                 0.0.0.0                 0             32768 ?
*>i172.16.2.0/30          10.255.255.5            0            100      0 ?
*>i172.16.3.0/30          10.255.255.6            0            100      0 ?
*>i172.16.3.0/24          10.255.255.6            0            100      0 65100 i
*> 192.168.1.0            172.16.1.2             0             0 65100 i
*>i192.168.2.0            10.255.255.5            0            100      0 65100 i
*>i192.168.3.0            10.255.255.6            0            100      0 65100 i
Route Distinguisher: 200:1 (default for vrf customerB)
*>i172.16.4.0/30          10.255.255.5            0            100      0 ?
*>i192.168.7.0            10.255.255.5            0            100      0 65100 i

```

PE-R4 ip bgp vpnv4 all

```

PE-R5(config)#do show ip route vrf customerA

Routing Table: customerA
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

    172.16.0.0/16 is variably subnetted, 4 subnets, 2 masks
B       172.16.1.0/30 [200/0] via 10.255.255.4, 02:15:37
C       172.16.2.0/30 is directly connected, FastEthernet0/1
B       172.16.3.0/30 [200/0] via 10.255.255.6, 02:25:34
B       172.16.3.0/24 [200/0] via 10.255.255.6, 02:25:34
B      192.168.1.0/24 [200/0] via 10.255.255.4, 02:15:37
B      192.168.2.0/24 [20/0] via 172.16.2.2, 02:25:37
B      192.168.3.0/24 [200/0] via 10.255.255.6, 02:25:34

```

PE-R5 ip route vrf customerA

```

PE-R5(config)#do show ip route vrf customerB

Routing Table: customerB
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

    172.16.0.0/30 is subnetted, 1 subnets
C       172.16.4.0 is directly connected, FastEthernet1/0
B       192.168.7.0/24 [20/0] via 172.16.4.2, 02:26:53

```

PE-R5 ip route vrf customerB

```

CE-CustA-2(config)#do show ip bgp summary
BGP router identifier 192.168.2.1, local AS number 65100
BGP table version is 7, main routing table version 7
4 network entries using 468 bytes of memory
5 path entries using 260 bytes of memory
4/2 BGP path/bestpath attribute entries using 496 bytes of memory
1 BGP AS-PATH entries using 24 bytes of memory
0 BGP route-map cache entries using 0 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
BGP using 1248 total bytes of memory
BGP activity 5/1 prefixes, 6/1 paths, scan interval 60 secs

Neighbor      V    AS MsgRcvd MsgSent   TblVer  InQ OutQ Up/Down  State/PfxRcd
172.16.2.1     4   400    161    153       7    0    0 02:29:27      3

```

CE-CustA-2 ip bgp summary

```

PE-R6(config)#do show ip bgp vpnv4 all summary
BGP router identifier 10.255.255.6, local AS number 400
BGP table version is 17, main routing table version 17
7 network entries using 959 bytes of memory
7 path entries using 476 bytes of memory
6/4 BGP path/bestpath attribute entries using 744 bytes of memory
1 BGP AS-PATH entries using 24 bytes of memory
1 BGP extended community entries using 24 bytes of memory
0 BGP route-map cache entries using 0 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
BGP using 2227 total bytes of memory
BGP activity 7/0 prefixes, 7/0 paths, scan interval 15 secs

Neighbor      V    AS MsgRcvd MsgSent   TblVer  InQ OutQ Up/Down  State/PfxRcd
10.255.255.4   4   400    156    156       17    0    0 02:30:15      2
10.255.255.5   4   400    159    156       17    0    0 02:30:42      2
172.16.3.2     4 65100    154    162       17    0    0 02:30:39      2

```

PE-R6 ip bgp vpnv4 all summary

```
PE-R6(config)#do ping vrf customerA 172.16.1.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 172.16.1.2, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 96/120/176 ms
PE-R6(config)#do trace vrf customerA 172.16.1.2

Type escape sequence to abort.
Tracing the route to 172.16.1.2

  1 172.16.1.1 [MPLS: Label 25 Exp 0] 64 msec 100 msec 88 msec
  2 172.16.1.2 116 msec 128 msec 144 msec
```

PE-R6 ping/trace vrf customerA 172.16.1.2

```
CE-CustA-1(config)#do ping 192.168.2.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.2.1, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 128/176/208 ms
CE-CustA-1(config)#do trace 192.168.2.1

Type escape sequence to abort.
Tracing the route to 192.168.2.1

  1 172.16.1.1 40 msec 36 msec 28 msec
  2 172.16.2.1 [AS 400] [MPLS: Label 26 Exp 0] 128 msec 144 msec 180 msec
  3 172.16.2.2 [AS 400] 188 msec 144 msec 236 msec
CE-CustA-1(config)#do ping 192.168.7.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.7.1, timeout is 2 seconds:
UUUUU
Success rate is 0 percent (0/5)
CE-CustA-1(config)#do trace 192.168.7.1

Type escape sequence to abort.
Tracing the route to 192.168.7.1

  1 172.16.1.1 28 msec 24 msec 36 msec
  2 172.16.1.1 !H  !H  !H
```

CE-CustA-1 ping/trace 192.168.2.1 (CE-CustA-2) (Si llega por mismo customer)

CE-CustA-1 ping/trace 192.168.7.1 (CE-CustB-2) (No llega por diferente customer)

```
R3(config)#service password-encryption
R3(config)#enable secret cisco123
R3(config)#username admin privilege 15 secret admin
R3(config)#line vty 0 4
R3(config-line)#
R3(config-line)#login local
```

R3 service password-encryption configuration

```
enable secret 5 $1$T9L5$CKGMqyi9JPvxRZwDdcssk0
```

```
username admin privilege 15 secret 5 $1$VcX1$nxVppee0Bmdbf2DRo9WjD1
```

R3 password and username encrypted

## 6. Conclusions

Milton

This project pushed me to combine several advanced networking concepts into a single, functional MPLS VPN design. From setting up EIGRP for internal convergence to configuring VRFs and MP-BGP sessions across the provider edge, every step required careful attention to detail. One of the most technically demanding parts was making sure the route-targets matched properly and that each customer's traffic remained completely isolated. The use of MPLS labels added a layer of complexity, especially when troubleshooting connectivity between routers.

Throughout the process, we encountered several configuration issues. Some routing problems were caused by incorrect network statements or missing route advertisements in BGP. We also faced ping failures between some routers, which were eventually traced back to mismatched subnets and VRF assignments. Additionally, a simple but impactful mistake was forgetting to secure the router with the correct enable secret password, which initially prevented us from testing the security requirements properly. These errors forced us to go back, review the configs carefully, and better understand how each part of the system interacts.

More than just a lab exercise, this practice showed me how fragile and precise network setups are in real environments. It reminded me that successful network engineering isn't just about knowing commands—it's about understanding behavior, testing results, and adapting when things don't go as planned.

### **Challenges**

Understanding the interaction between VPNv4 address-family configuration and VRF route-targets was the most complex part. Ensuring proper propagation of labels and peerings also required close attention to interface IPs, loopback sources, and MPLS enablement.

## **7. References**

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